

Problem Set 2 — Recurrences & the Master Theorem

VCE Algorithmics (HESS) · Unit 4 AOS1 · Week 3

Due: start of Week 4 (Memo 02 workshop). Question E2 feeds directly into your Memo 02.

Name: _____

Due: Monday 3 August

Method reminder: read $T(n)$ off the code $\rightarrow$ solve by expansion, level-sums, or Master Theorem $\rightarrow$ check the form applies before using the theorem $\rightarrow$ name your tool in the justification.

Section A — Writing recurrences

For each algorithm, write the recurrence for $T(n)$ including the base case. *Do not solve yet.*

A1.

```
1 Algorithm Echo(n):  
2   If n = 0 Do  
3     Return  
4   print(n)  
5   Echo(n - 1)
```

A2.

```
1 Algorithm SplitScan(L):           # n = length(L)  
2   If n = 1 Do  
3     Return  
4   Foreach x in L Do  
5     print(x)  
6   SplitScan(first half of L)  
7   SplitScan(second half of L)
```

A3.

```
1 Algorithm Corridor(n):  
2   If n = 1 Do  
3     Return 1  
4   Return Corridor(n - 1) + Corridor(n - 1)
```

Section B — Solving by expansion

B1. Solve $T(n) = T(n - 1) + c$, $T(0) = c$, by expanding three steps, stating the pattern, and closing it off. Give the final Θ bound.

B2. Solve $T(n) = T(n/2) + c$, $T(1) = c$, the same way.

B3. Solve $T(n) = T(n-1) + cn$, $T(1) = c$. *Hint: the work terms form a familiar sum.*

Section C — Master Theorem

For each recurrence: check the form applies, identify a , b , $f(n)$, compute $n^{\log_b a}$, name the case, and state the result. (2 marks each style — form + case + result.)

C1. $T(n) = 2T(n/2) + n$

C2. $T(n) = 8T(n/2) + n^2$

C3. $T(n) = T(n/3) + 1$

C4. $T(n) = 2T(n/2) + n^3$

C5. $T(n) = 3T(n/9) + n$

C6. $T(n) = 2T(n-1) + 1$ (*careful!*)

Section D — Mixed analysis

D1. State the worst-case complexity of `SplitScan` (A2), using your recurrence from Section A. Solve it with whichever tool applies, and name the tool.

D2. A student solves $T(n) = 4T(n/2) + n$ with Master Theorem case 3 and answers $\Theta(n)$. Identify the error, and give the correct case and bound.

D3. Memoised Fibonacci stores every computed value. Explain, using the call graph, why memoisation collapses $\Theta(\phi^n)$ to $\Theta(n)$.

Section E — Challenge and SAT bridge

E1. Towers of Hanoi: $T(n) = 2T(n-1) + 1$, $T(1) = 1$. Show by expansion that $T(n) = 2^n - 1$ exactly, then verify for $n = 3$.

E2. *Memo 02 rehearsal.* Choose one module of your SAT solution (recursive if you have one; otherwise the module with the deepest loop nesting). Write: (i) its recurrence or loop structure, (ii) the tool you used, (iii) the tightest worst-case bound, (iv) one sentence on what this means for your solution at $10\times$ the input size.

We will mark Sections B and C together at the start of the Week 4 workshop.